Fundamental Frequency Extraction Using DCT Based Cumulative Power Spectrum in Noisy Speech

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Abstract

This research proposes an effective concept specifically designed for speech processing applications, particularly in the retrieval of precise pitch from speech signals under noisy conditions .

According to the experimental results, our proposed technique demonstrates superior performance in noisy conditions compared to other existing state-of-the-art methods such as PEFAC, BaNa, and YIN without utilizing any kind of post-processing techniques in our proposed method.

Objectives

Our project is all about improving how we detect pitch of speech signal, especially when there's a lot of background noise.

To achieve this goal, we introduce a fundamental frequency extraction algorithm designed to tolerate non-stationary changes in the amplitude and frequency of the input signal.

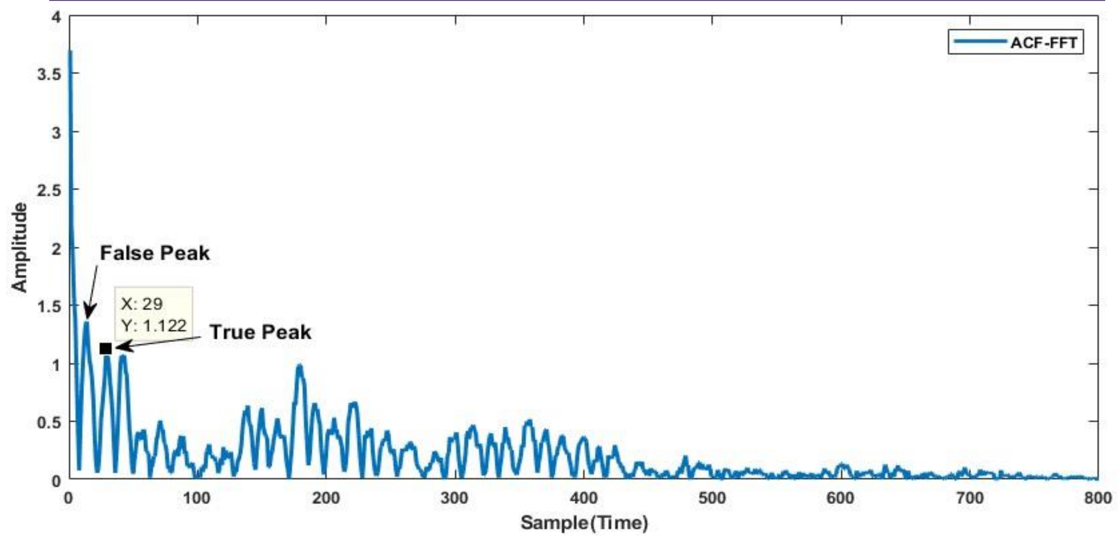
To enhance the accuracy of fundamental frequency extraction, our focus is on preserving the original state of speech harmonics while suppressing noise elements inherent in noisy speech signals.
Design a pitch extraction algorithm which is:

- Simple and efficient.
- Time saving.
- Robust against both noise and vocal tract effect.
- Without relying on post-processing

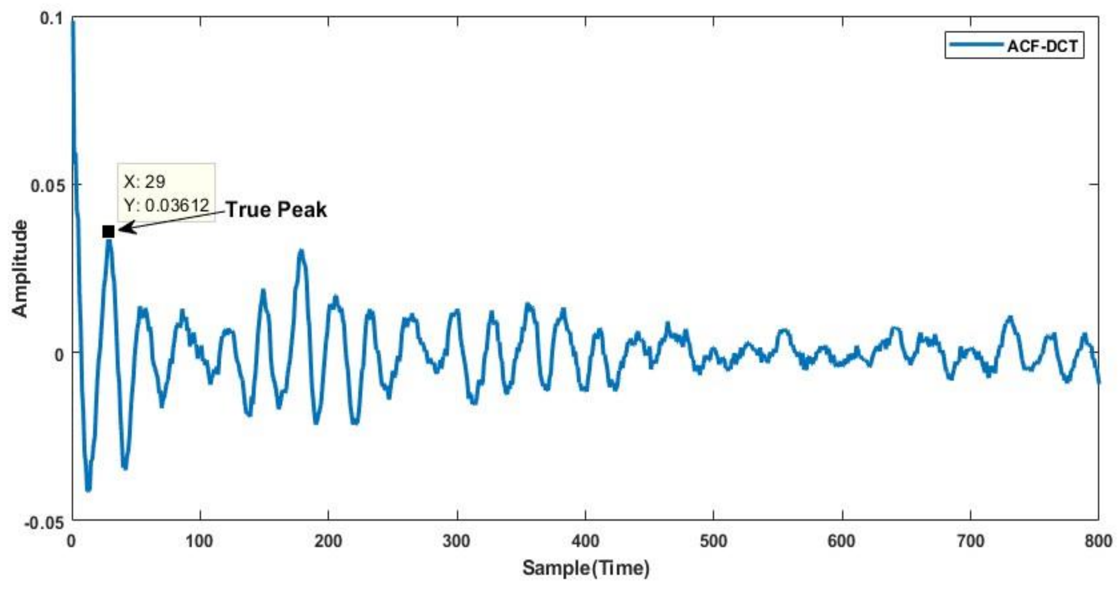
Motivation

Most existing methods struggle with accuracy in noisy conditions and it gets even tougher when the SNR is low.
Pitch detection in noisy environments within low SNR, when dealing with a real world noise, has only been attempted by a few researchers. We're taking a fresh approach by exploring the potential of using DCT instead of the usual FFT in ACF.
The proposed method isn't just about overcoming noise issues in pitch detection, it also overcomes the struggling of other methods within low SNR. We're aiming to make a contribution that can make pitch detection better and more reliable, especially in real-world situations .

Problem Formulation



The conventional ACF output waveform is impacted by the vocal tract effect, resulting in a False Peak close to the True Peak .



The adoption of DCT in place of FFT within ACF helps alleviate the vocal tract effect. DCT based ACF gives true peak at 29 but error is present because of noise.

Proposed Method

We employ a DCT based cumulative power spectrum, as opposed to a power spectrum by supporting the ACF based approach to enhance the overall performance of the ACF. We enhance our method by utilizing shorter sub-frames of the input signal to mitigate the noise characteristics present in speech signals.

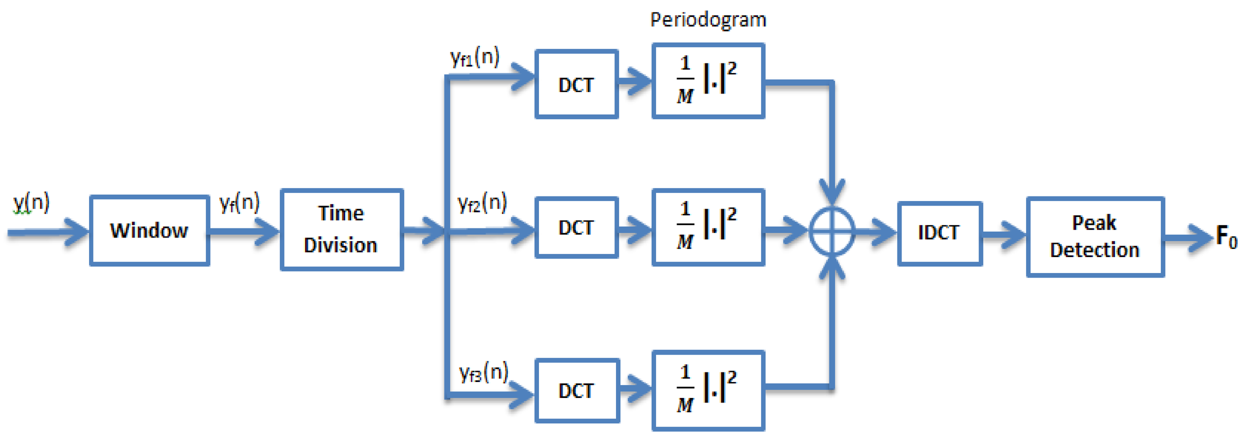
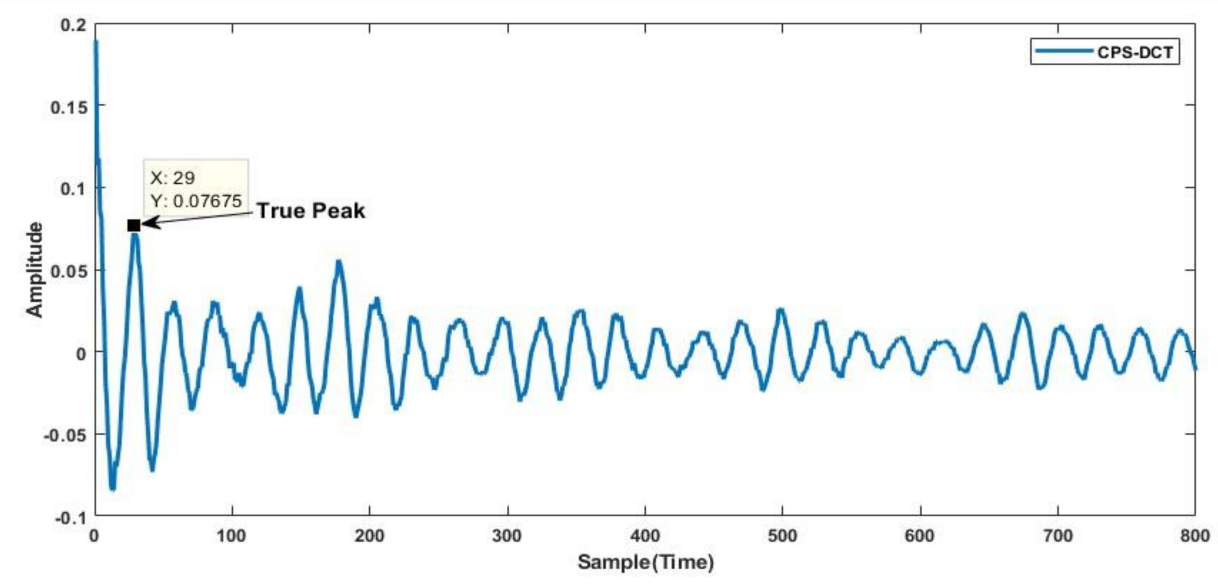


Fig: Block Diagram of Cumulative Power Spectrum

Observation of Our Proposed Method



Our proposed method plays a crucial role in achieving a smoother signal than the DCT-based ACF.
It not only significantly reduces the vocal tract effect but also provides a more seamless waveform compared to other methods.

Results From Experiments

Experimental Conditions

- Frame Length : 50 [*ms*] without PEFAC and BaNa.
- Frame shift : 10 [*ms*].
- Window Function : Rectangular Window except for PEFAC and BaNa.
- DCT(IDCT) points : 2048 points without PEFAC and BaNa.

Average GPE Comparison for KEELE Database for Male & Female

White Noise

SNR	CPS-DCT(Proposed)	PEFAC	BaNa	YIN
-5	29.55031	41.04161	27.44827	48.99407
0	20.58926	37.15398	22.61711	31.3701
5	15.9896	34.38029	19.58053	21.59396
10	13.86573	33.01514	17.80155	16.57429
15	13.12183	32.50849	16.97529	14.29189
20	12.90138	31.98937	16.59966	12.8749

Babble Noise

SNR	CPS-DCT(Proposed)	PEFAC	BaNa	YIN
-5	52.91545	58.59012	55.15213	52.5633
0	35.182864	49.01548	40.54781	36.8909
5	22.888482	41.86631	29.48598	23.68052
10	16.577184	37.4189	22.84607	16.64691
15	13.09545	34.98688	19.69209	13.1676
20	11.875954	33.395	17.70592	12.14275

Train Noise

SNR	CPS-DCT(Proposed)	PEFAC	BaNa	YIN
-5	49.12419	50.73216	39.07639	50.23369
0	33.44969	43.17706	29.08431	34.38243
5	22.81971	38.9921	23.11176	22.76062
10	16.98957	35.59503	20.04322	16.36671
15	14.50046	33.40132	18.31063	13.42459
20	13.49092	32.25406	17.36339	12.16158

HF-Channel Noise

SNR	CPS-DCT(Proposed)	PEFAC	BaNa	YIN
-5	38.53928	45.69573	27.1985	49.37144
0	24.70326	40.13406	22.64961	31.55247
5	17.64917	36.86074	19.82379	21.01063
10	14.79154	34.37672	17.90747	16.06053
15	13.45732	32.98622	17.31619	13.76763
20	13.04867	32.11695	16.57342	12.79615

Discussion

Based on the findings from these tables, our suggested approach consistently demonstrates the lowest average GPE rate among all compared techniques across practically all SNRs in presence of Babble noise.

Conclusion

We introduce an improved method that excels in isolating noise from the waveform, particularly in Babble noise scenarios, outperforming other techniques. This method exhibits a lower average GPE rate compared to alternative approaches without any complicated post-processing. It efficiently mitigates the impact of vocal tract effects by equalizing unnecessary ripples in the waveform .

References

[1] Savitha S Upadhya. Pitch detection in time and frequency domain. In *2012 International Conference on Communication, Information & Computing Technology (ICCICT)*, pages 1–5. IEEE, 2012.
[2] Md Saifur Rahman. Pitch extraction for speech signals in noisy environments.*(No Title)*, 2020.
[3] Celia Shahnaz, Wei-Ping Zhu, and M Omair Ahmad. Pitch estimation based on a harmonic sinusoidal autocorrelation model and a time-domain matching scheme. *IEEE transactions on audio, speech, and language processing*, 20(1):322–335, 2011.

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